Russell's Paradox

1. $\emptyset \notin \emptyset$, $N \in \{N\}$, $Z \in \{Z\}$,

$Z \notin Z$. $\{Z\} \in \{N, \{Z\}\}$,

but $\{Z\} \notin \{Z\}$

1.3 x is set $\Rightarrow x \notin x$

2. $R = \{x \mid x \text{ is set}\}$

$\Rightarrow$ collection R is not a set.

proof: If R is set, $\Rightarrow R \in R$,

error (1.3)

$\therefore$ R is not set.

3. $R = \{x \mid x \in S, x \text{ is set}\}$

$\Rightarrow R$ is set.

(need proof) ε

4. If P is any property

$\not\Rightarrow R = \{x \mid P(x)\}$ is a set.

(from 2)

4.3 we can't form a set by simply collecting all objects with a given property.

4.4 Not every collection we want to be a set actually is a set.

5. $\forall x$, $\langle x \rangle$ is a set generated by x.

$R = \{x \mid x \notin \langle x \rangle\}$

$\Rightarrow \nexists r$, s.t., $\langle r \rangle = R$.

proof: If $\exists r$, s.t., $\langle r \rangle = R$.

Then, if $r \in \langle r \rangle \Rightarrow r \notin R = \langle r \rangle$

if $r \notin \langle r \rangle \Rightarrow r \in R = \langle r \rangle$

$\therefore r \in \langle r \rangle \Leftrightarrow r \notin \langle r \rangle$,

error. $\therefore r \nexists$.

6. Russell's Paradox

$R = \{x \mid x \text{ is set}, x \notin x\}$

$\Rightarrow R$ is not set.

proof: 6.3 method 1; (modern set theory)

$\because R = \{x \mid x \text{ is set}\}$ $\therefore \nexists$ set R

(from 2)

6.4 method 2. (naive set theory)

(allow $x \in x$)

If R is set, Then $R \in R$ or $R \notin R$.

If $R \in R \Rightarrow R \notin R$.

If $R \notin R \Rightarrow R \in R$

$\therefore R \in R \Leftrightarrow R \notin R$, error. $\therefore$ set $R \nexists$.

7. Barber Paradox.

Barber b: $\forall x \in U$, $S(b,x) \Leftrightarrow \text{not } S(x,x)$

$\Rightarrow b \nexists$. here $S(x,y) := x$ shaves y, $\forall x, y \in U$

U = set of all men in the village.

proof: let $\langle x \rangle$ be set $\{y \in U \mid S(x,y)\}$

$\therefore B = \{x \mid x \in U, x \notin \langle x \rangle\}$

$\Rightarrow \nexists$ barber b, s.t., $\langle b \rangle = B$ (from 5)